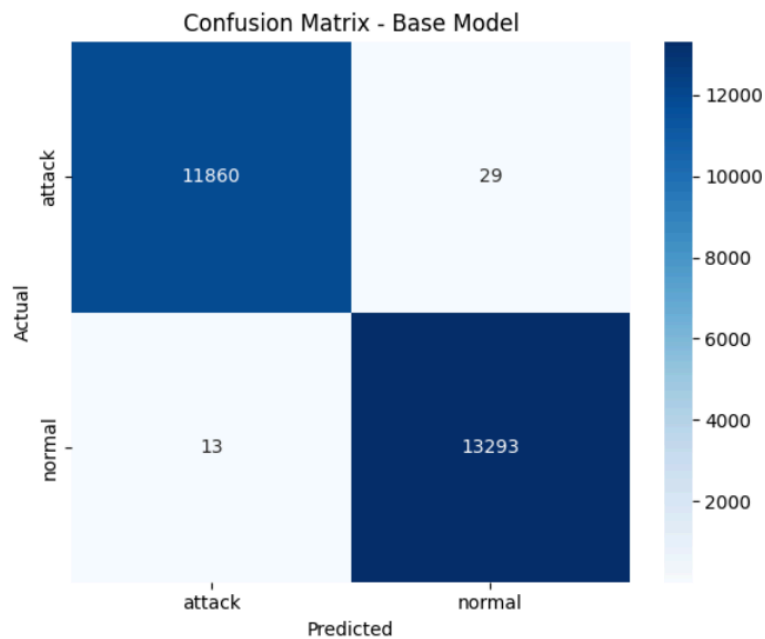


3. c) Evaluating base model on test set

Classification Report (Base Model):

	precision	recall	f1-score	support
attack	1.00	1.00	1.00	11889
normal	1.00	1.00	1.00	13306
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Confusion Matrix (Base Model):



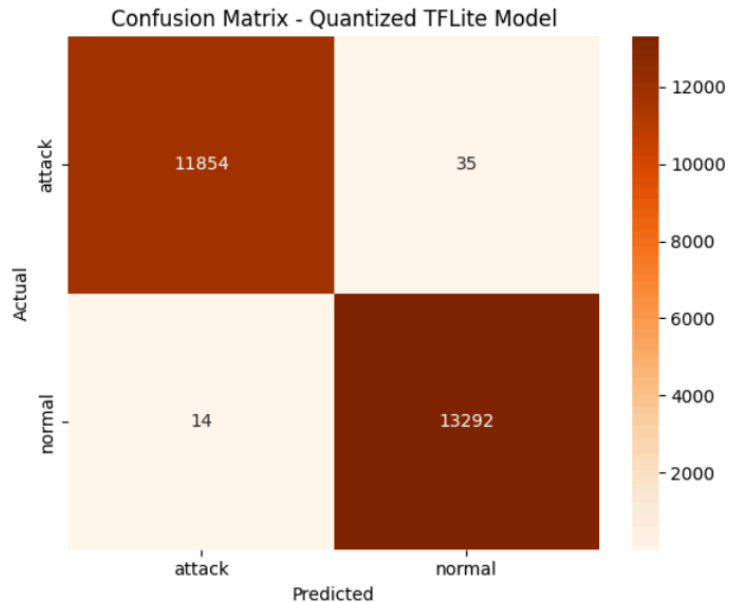
Based on the classification report and confusion matrix, the base model performs really well on the test set. Accuracy for both “attack” and “normal” are high (all are 1.0). We can also see in the confusion matrix that the number of incorrect predictions are relatively low (29 and 13) compared to the correct predictions (11860 and 13293).

4. b) Evaluating tflite_quant_model on test set

Classification Report (Quantized TFLite Model):

	precision	recall	f1-score	support
attack	1.00	1.00	1.00	11889
normal	1.00	1.00	1.00	13306
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Confusion Matrix (Quantized TFLite Model):



Based on the classification report and confusion matrix, the quantized model performs really well on the test set. Accuracy for both “attack” and “normal” are high (all are 1.0). We can also see in the confusion matrix that the number of incorrect predictions are relatively low (35 and 14) compared to the correct predictions (11854 and 13292). These results illustrate that our model quantization performed well, as we were able to compress the model while keeping the same accuracy.

5. c) Serial Monitor outputs for five samples:

```
Start setup: Model loaded OK
Ops registered
Interpreter created
AllocateTensors() OK
Input type: 72
Output type: 72
Input size: 15777
Sample #1 | Predicted Class: normal | Actual Class: normal
Sample #2 | Predicted Class: normal | Actual Class: normal
Sample #3 | Predicted Class: attack | Actual Class: attack
Sample #4 | Predicted Class: attack | Actual Class: attack
Sample #5 | Predicted Class: normal | Actual Class: normal
```

d) Serial Monitor outputs for ten samples:

Start setup: Model loaded OK

Ops registered

Interpreter created

AllocateTensors() OK

Input type: 72

Output type: 72

Input size: 15777

Sample #1		Predicted Class: normal		Actual Class: normal
Sample #2		Predicted Class: attack		Actual Class: normal
Sample #3		Predicted Class: attack		Actual Class: attack
Sample #4		Predicted Class: normal		Actual Class: attack
Sample #5		Predicted Class: normal		Actual Class: normal
Sample #6		Predicted Class: normal		Actual Class: normal
Sample #7		Predicted Class: normal		Actual Class: attack
Sample #8		Predicted Class: normal		Actual Class: attack
Sample #9		Predicted Class: normal		Actual Class: normal
Sample #10		Predicted Class: normal		Actual Class: normal